Certified Rewrite Discovery for Tensor Programs: A Machine-Minted Conv+BN Fold from Minting to Runtime

Draft — proposed, not accepted

mlopt project

2026-10-20

Abstract

Compilers choose better implementations only from transformations a human already wrote. We describe a pipeline in which the rewrite rules themselves are a product of measurement: laws are data, composites are minted by a fixpoint over the shipped library, guards are transported through an evidence store, derivability is credited against an object's own premise closure, and admission is decided by a staged gauntlet that has repeatedly falsified the engine's own candidates. The flagship result is `conv_bn_fold` — the eval-mode batch-norm weight fold a hand optimizer writes — minted, admitted, and shipped verbatim as a machine-authored law in the default rule set. The deployed program evaluates bitwise-equal to the spelled one through an independent runtime and is 16% cheaper per call (72.8 vs 87.0 ms), with the term-level cost model agreeing in direction. All reported numbers are reproducible from the committed tree. The scope is deliberately honest: a kernel-scale, certified-rewrite result, not a model-scale speedup claim.

> **!** Important
>
> **Status.** This is a working draft — *proposed, not accepted*. It has not had human revision or external review. Every quantitative claim names the probe, test, or retro that produced it; the rest of the paper should be read as a candidate for rewriting, not as a settled result.

1 Introduction

Every optimization a production compiler knows was written by a person — a hand-derived rule, encoded per framework. The rule library is the bottleneck: human authorship is slow, and the pairings nobody names never get found. For a machine to *author* optimizations, four things must exist: rules it can write (data, not code), a referee to check them, a score to rank them, and a world — a runtime — that programs, rules, and derivations all live inside.

The pieces have arrived separately. Massalin's superoptimizer (Massalin 1987) searched programs exhaustively; equality saturation (Tate et al. 2009; Willsey et al. 2021) closes the program space under a fixed law bank, and TENSAT applies it to tensor graphs. Rule-inference work (Ruler (Nandi et al. 2021), the Babble line (Cao et al. 2023)) grows the library itself. And learned play on well-defined games — AlphaGo Zero (Silver et al. 2017), AlphaTensor (Fawzi et al. 2022) — reaches structure beyond human-written heuristics. The synthesis story for *tensor programs with runtime-priced deployment*, though, is thinner: no system connects discovery, certification, pricing, shipping, and learned guidance inside one substrate.

This paper reports the connection — and one complete measured traversal of it — in the mlopt engine. The contribution is not any single mechanism but their composition: a categorical substrate makes rules data, derivations checkable, cost measurable, and provides the runtime world — which closes discovery into a well-defined game played by learned policies. The enabling property is *self-containedness*: programs, rewrite laws, and derivations are n-morphisms in one calculus — executable values, not external metadata — so the system produces the values it consumes. To our knowledge this is the first *operational* substrate of this kind; categorical languages prior to it are denotational or design-level, not working machines whose runtime executes the morphism hierarchy. The claim is narrow and checkable:

> A rewrite a hand optimizer would write — the eval-mode conv+BN weight fold — was **discovered, certified, priced, and shipped by the machine**, and its deployed form runs bitwise-equal and cheaper through an independent runtime.

No model-scale speedup is claimed. The result is kernel-scale (one op pair, fp64 fixtures), the runtime is a tree-walk interpreter (debug build), and the move alphabet is bounded by an 18-op host-tensor surface — all named in Section 6. What the case study establishes is the *pipeline*: that a law can travel from measured need to a shipped, certified, runtime-priced rule without a human writing it.

The repository is `handl-lang/research-mlopt`; every number below cites the artifact that produced it (§Section 7).

2 Background

2.1 Equality saturation

Given a term and a set of rewrite laws, an e-graph compactly represents the closure: all programs reachable by the laws, sharing subterms, until a fixed point (Tate et al. 2009). Extraction then picks a cheapest member under a cost model. The soundness boundary is the law bank: any unsound rule poisons every e-class it touches, so in mlopt a rule's side conditions (`check`/`derive`/declarative `cond`) are the *only* layer that keeps a false member out — the certificate replay re-proves that a step is a valid application, not that the rule is true.

2.2 Laws as data on a categorical substrate

Catopt reads a program as a 1-morphism between tensor types, a rewrite law as a 2-cell between programs, and a derivation — one law proven from premises — as a 3-cell. The consequence that matters operationally: laws are **data rows** (an LHS/RHS pattern pair plus a guard spec), serializable, storable, composable. The shipped bank (`mlopt_core.laws.ALL_RULES`, 84 rules) mixes human-authored axioms with twelve machine-admitted objects (`mlopt_core.laws.provenance.MACHINE_SHIPPED`).

2.3 The handl substrate and mlopt-kit

The same "one language, many interpretations" model is implemented by the handl platform: eval, typecheck and rewrite are interpretations of one language whose tables are data. `packages/mlopt-kit` — a plain Rust crate — composes handl's canonical `Language`, its lex/parse/tree-walk-exec engines, and `handl-host-tensor`'s 18-op `tensor.*` surface as dependencies (no forks). `hnl-run` is a subprocess door: `.hnl` in, a JSON wire tensor out, `--repeat N` evaluating a source N times in one process — the measurement hook. `lab/kit/term_hnl.py` renders any mlopt IR term as a `.hnl` program from a data table (per-op arg assembly, attr marshaling, leaf binding), so the *same object the e-graph rewrites* is what the runtime executes.

3 Machinery

3.1 The mint fixpoint

`mlopt_core.meta.synthesize_rules` runs a bounded fixpoint over the law bank on a seed term: each round applies rules, composes the pairs that chain, and emits `syn_*` composites with provenance — the names of the premise rules each composite was built from. On the eval-mode ConvBN seed it emits 974 rules in 12 s (fuel 30,000, 4 rounds); the fold composite mints deterministically (identical names across runs).

3.2 Guard transport

A minted rule's side conditions are closures over its premise objects, so the evidence store keeps them as *specs*, not code: each stored record carries a `"guard"` field — parent names plus re-expression maps — and `evidence.admit_object` rebuilds `check`/`derive` by resolving parents as store rows, transitively. The shipped form (`mlopt_core.laws.minted`) embeds the stored-closure records as data and replays the same rebuild semantics at import — louder than the store: a parent resolving short of declared strength raises rather than admitting a degraded rule.

3.3 Premise-anchored derivability

`verify_law` answers "is this instance derivable under a rule bank?" with a replayable witness. For a stored composite the honest question is *"derivable under the premises it was minted from"* — so the derivability leg evaluates against `base_rules` plus the object's transitive premise closure, with an anti-gaming bar: the certificate must be standalone-replayable and every step naming a non-axiom rule must *site-verify* under the fp64 numeric oracle. A premise false at its applied site revokes the credit and the verdict falls back to the axioms-only answer.

3.4 One cost alphabet

The cost model credits compile-time work structurally: a subtree closed over `Param`/`Const` leaves whose root op is in `FOLD_OPS` — the 161-op fold-at-lowering alphabet, single-sourced between the lowerer, the cost predicate and the extractor — is priced at ~0 marginal. This is what lets a fold's pay leg measure what it should: the fold's `mul`/`reshape`/`rsqrt` chain is once-per-deployment work, not per-call work, and `dag_cost` prices it that way.

3.5 The admission gauntlet

`evidence.run_gauntlet` stages the verdict: reconstruct → full-data → measure → truth → novelty → typed-pay → closure → certificate. Truth is two legs — the numeric oracle over the guarded region *and* premise-anchored derivability; novelty refuses re-spellings of shipped laws (`relation=duplicate`); typed-pay suppresses cost drops that mint ill-typed members; closure bounds the firing footprint. The gauntlet has repeatedly falsified the engine's own candidates — a naive broadcast guard with 212 bad accepts, an apparent paying sibling with semantically unequal sites, four deliberate false twins (`project/retros/mul-unsq-nat.md`, `layer3-loop.md`, `typed-pay-gate.md`).

3.6 Learned guidance as a game

Because laws are data, discovery admits a well-defined game structure: rule applications and compositions are well-formed moves; the admission gauntlet is the referee — a move only scores if its artifact

survives admission; the cost alphabet is the score. This is implemented machinery, not aspiration: `engine` defines the `Board` protocol and episode/train/eval drivers, `players` a `LinearPolicy` arm over a domain-specified featurizer, and `play` a domain registry (`meta`, `joint`, `search`, `torch`, `gen`, `open`; `torch` and `gen` ship `deliver` hooks that lower the winning extraction to a verified module). A domain is data — a case generator, a board factory, and a feature schema — not a subclass.

Two linear priors ship with pinned held-out wins: the open-game policy ranks law candidates at ~4.7× the cost-delta heuristic's average precision (AP 0.210 vs 0.045) and ~2× useful laws per episode, and the meta-arena prior beats the scripted playbook (mean episode reward −0.06 vs −0.10) — the latter flagged by its own eval as possibly sampling-variance-riding, an honest negative kept in the record. A distilled contraction policy (`PairPolicyNet`) beats `opt_einsum`'s randomized greedy (0.91× cost at n=40) and ships as a standalone artifact. Conversely, a neural rule-value policy *loses* to the heuristic (AP 0.023 corpus-trained; 0.041 parity when trained on-distribution — a data gap, not capacity), and two learned-policy axes (schedule guidance; direct law invention) were measured and falsified — the evidence for learned guidance is real but early (controlled baseline and ablations in `docs/results_learned_baseline.md` / `docs/results_learned_ablations.md`): linear and distilled arms beat baselines; neural methods do not yet.

4 Case study: the conv+BN fold

The rewrite is the standard eval-mode fold:

$$\mathrm{BN}_{\gamma,\beta,\mu,\sigma^2,\varepsilon}(\mathrm{conv}(x,W,b)) \;=\; \mathrm{conv}(x,\; W \odot \hat{s}',\; b \odot s + (\beta - s \odot \mu)), \qquad s = \gamma \cdot (\sigma^2+\varepsilon)^{-1/2}.$$

The account below is the probe's own output (`lab/probes/conv_fold_premise_pay_probe.py`, ~13 s).

Anchor. `CONV_FOLD_RULES` provides the channel-scale and channel-shift anchors (`mul(conv2d(x,W,b),ŝ)`-shaped single-step folds); `BATCH_NORM_RULES` expands `batch_norm` into its affine spelling. Individually they are enabled-but-opt-in — the anchors refuse at truth-starvation, the expand at typed-pay — their value is *enablement*: they are the fold's derivation premises.

Mint. `synthesize_rules` over the bank minus the already-shipped composite emits 974 rules; the fold composite — `batch_norm(conv2d(·),…)` → `conv2d(·)` — mints as a depth-4 `syn_*` chain over an 8-rule transitive premise closure (five shipped anchors + three intermediates).

Transport and truth. Stored with its closure, `admit_object` rebuilds both hooks from guard records (`missing=('check','derive')` → transported). The transported guard accepts the real site (1 eq / 0 ne) and 105/105 of the guarded synth region with 255 honest declines; `verify_law` replays the derivation under shipped axioms in one step — `witness=('conv_bn_fold',)`, the library law proving its own minted origin.

Pay. On the 8×8 probe site the folded instance drops `dag_cost` **34,874.3 → 17,438.7** (−50 %), fires once, verifies `pass` on the lowered modules.

Ship. The composite is promoted *verbatim*: its stored records embed in `mlopt_core.laws.minted`, its guard re-materializes at import, `MACHINE_SHIPPED` grows to 12. Re-running the gauntlet post-ship refuses at novelty with `relation=duplicate` — the documented signature of "already admitted."

Runtime-verified. Through the kit bridge (Section 2), the spelled term `batch_norm(conv2d(x,W,b),…)` and the deployed term `conv2d(x,W2,b2)` — with `W2,b2` themselves mlopt subterms materialized through the kit — evaluate to the same tensor (**bitwise-equal** on the seeded fixture; the full `torch.export` IR of the module evaluates at 6.7e-16 max abs diff), and per-call wall over `hnl-run --repeat 30` measures three regimes — the honest deployed arm (fold math amortized to load) is ∼−5 %; an inline-fold arm that recomputes the fold per call runs *slower* (+22–33 %); and a subprocess-arm −16 % figure conflates the prelude's unused fold-marshal work into the baseline. `dag_cost` is the stable headline: 55,296 → 36,864 (−33 %). Full reconciliation: `bench/RESULTS.md` + `lab/probes/conv_fold_wall_regimes_probe.py`.

: The fold decision, priced by the runtime itself — honest regime-split. {#tbl-decision}

One honest subtlety: `dag_cost`'s param-only discount models compile-time folding the tree-walk runtime does not perform, so the measured arm separates fold-once (materialize `W2,b2`) from run-every-call (the deployed conv). The cost model and the wall agree in direction; the wall numbers are debug-build timings — directional, not a benchmark rig.

5 Related work

Equality saturation and e-graph superoptimization. Tate et al. (Tate et al. 2009) introduced the technique; egg (Willsey et al. 2021) made it practical; TENSAT (Yang et al. 2021) and follow-ons apply it to tensor graphs with fixed, hand-written rule sets. Catopt uses the same substrate but the rule bank is not fixed — the discovery loop writes into it through the gauntlet. egglog (Zhang et al. 2023) serves as a test-only differential oracle for the hand-rolled saturation.

Rewrite-rule inference and library learning. Ruler (Nandi et al. 2021) infers rewrite rules for scalar domains by enumeration + equality-saturation filtering — the direct ancestor of the mint-then-verify shape here, differing in that mlopt mints *composites over tensor laws* with transported guards and credits derivability against an object's own premise closure. Babble (Cao et al. 2023) learns library abstractions by anti-unification; mlopt's definitional-atom layer is the same move and measured **null** on reach-expansion — the paying layer is above it.

Generated rule banks. TASO (Jia et al. 2019) auto-generates graph substitutions for DNNs and verifies them formally with an automated theorem prover — the strongest prior on the verify-what-you-mint leg; the generated bank ships once and does not re-enter subsequent discovery. Isaria (Thomas and Bornholt 2024) synthesizes *sound* rewrite-rule systems from ISA specifications to drive e-graph vectorizing compilers, and Enumo (Pal et al. 2023) makes rule inference programmable — both grow the bank offline, and the ruleset a compiler runs is static. Glenside (Smith et al. 2021) is the closest precedent for machine-derived tensor transformations: e-graphs rediscovering access-pattern rewrites (im2col-style) — the shape of the conv+BN fold here.

Learned play over rewriting. CompilerGym (Cummins et al. 2022) casts compiler optimization as an RL environment over a fixed action space; the Cambridge line — Omelette (Singh 2022), MCTS-GEB (He, Singh, and Yoneki 2023), Aurora (Bărbulescu et al. 2024) — trains RL/MCTS policies that steer e-graph construction. Learned play over rewriting exists, but over a static bank. AlphaGeometry (Trinh et al. 2024) couples a learned proposer to a certifying symbolic deducer — learned play inside a verified loop, over a fixed deduction closure (AlphaTensor (Fawzi et al. 2022) similarly searches one fixed structure).

Each neighbor holds pieces of the loop — generation, certification, learned play — but, to our knowledge, none makes the minted bank itself an object of play: in mlopt a law minted during

discovery re-enters as a move inside the learned game (Section 3.6). The differentiation is that closure, not any single leg.

Compilers. TVM (Chen et al. 2018) autotunes schedules; TorchInductor (Ansel et al. 2024) fuses pointwise ops and codegen's the graph a human optimizer could reach. Both are incumbents in the repo's benchmark tables (wins and losses alike); the contribution here is not a faster kernel but a certified, machine-discovered *rewrite*.

Certified/verified compilation. The per-artifact boundary design — law, search, delivery, bound each with its own verifier — is in the spirit of translation validation; what is unusual is that the verifier also audits the *discovery* machinery (adversarial sweeps falsifying minted guards).

6 Limitations

- **Kernel scale.** One op pair, fp64 fixtures, a 1×4×24×24 activation. Nothing here measures model-scale compilation; wider-corpus evidence in the repo (bimodal exposure, ~1.5% yield of fresh paying structure) is honest that this is not yet a general speedup.
- **Move alphabet bound.** The kit surface is 18 `tensor.*` ops; the fold's own `s = ·rsqrt(²+)` is outside it (no elementwise `rsqrt`) and rides a marshaling channel. Emitted terms are confined to what the surface spells; outside-surface ops refuse by name.
- **Tree-walk runtime.** `hnl-run --repeat` timings are wall-clock on a debug-build tree-walk interpreter — they establish direction (the fold's value lands in the deployed program), not competitive throughput. No codegen claim is made.
- **Single-case gauntlet.** The admission measured here is a one-term corpus; the machinery's broader soundness evidence lives in the library-level sweeps, which caught real holes — but on different objects than this one.
- **Attr-narrow LHS.** The shipped composite's LHS carries the seed's literal attrs (`stride=(1,1)`, `padding=(1,1)`, `eps=1e-05`, `training=False`): its region is the default-attr eval-mode fold — honest, narrow, the common spelling.

7 Reproduction

All numbers are reproducible from the committed tree at `handl-lang/research-mlopt`:

```
uv sync
.venv/bin/python lab/probes/conv_fold_premise_pay_probe.py   # the admission account (~13 s)
.venv/bin/python -m pytest tests/test_kit_bridge.py -q       # the runtime-priced decision (8 t
cargo test -p mlopt-kit                                     # the Rust consumer (21 tests)
.venv/bin/python -m pytest -m "not slow"                     # the dev lane
```

The evidence ledger is `bench/RESULTS.md`; the measured retros each claim cites are under `project/retros/` (notably `minted-composite-default.md`, `kit-term-bridge.md`, `premise-anchored-derivabi fold-cost-single-source.md`); the extended working draft with the full win/loss tables is `docs/paper_draft.md`.

References

Ansel, Jason, Edward Yang, Horace He, Natalia Gimelshein, Animesh Jain, Michael Voznesensky, Bin Bao, et al. 2024. "PyTorch 2: Faster Machine Learning Through Dynamic Python Bytecode

Transformation and Graph Compilation." In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 929–47. ACM. https://doi.org/10.1145/3620665.3640366.

Bărbulescu, George-Octavian, Taiyi Wang, Zak Singh, and Eiko Yoneki. 2024. "Learned Graph Rewriting with Equality Saturation: A New Paradigm in Relational Query Rewrite and Beyond." *arXiv Preprint arXiv:2407.12794*.

Cao, David, Rose Kunkel, Chandrakana Nandi, Max Willsey, Zachary Tatlock, and Nadia Polikarpova. 2023. "Babble: Learning Better Abstractions with e-Graphs and Anti-Unification." *Proceedings of the ACM on Programming Languages* 7 (POPL): 396–424. https://doi.org/10.1145/3571207.

Chen, Tianqi, Thierry Moreau, Ziheng Jiang, Lianmin Zheng, Eddie Yan, Haichen Shen, Meghan Cowan, et al. 2018. "TVM: An Automated End-to-End Optimizing Compiler for Deep Learning." In *13th USENIX Symposium on Operating Systems Design and Implementation (OSDI)*, 578–94. USENIX Association.

Cummins, Chris, Bram Wasti, Jiadong Guo, Brandon Cui, Jason Ansel, Sahir Gomez, Somya Jain, et al. 2022. "CompilerGym: Robust, Performant Compiler Optimization Environments for AI Research." In *Proceedings of the 20th IEEE/ACM International Symposium on Code Generation and Optimization (CGO)*, 92–105. IEEE. https://doi.org/10.1109/CGO53902.2022.9741258.

Fawzi, Alhussein, Matej Balog, Aja Huang, Thomas Hubert, Bernardino Romera-Paredes, Mohammadamin Barekatain, Alexander Novikov, et al. 2022. "Discovering Faster Matrix Multiplication Algorithms with Reinforcement Learning." *Nature* 610: 47–53.

He, Guoliang, Zak Singh, and Eiko Yoneki. 2023. "MCTS-GEB: Monte Carlo Tree Search Is a Good e-Graph Builder." In *Proceedings of the 3rd Workshop on Machine Learning and Systems (EuroMLSys)*. ACM. https://doi.org/10.1145/3578356.3592577.

Jia, Zhihao, Oded Padon, James Thomas, Todd Warszawski, Matei Zaharia, and Alex Aiken. 2019. "TASO: Optimizing Deep Learning Computation with Automatic Generation of Graph Substitutions." In *Proceedings of the 27th ACM Symposium on Operating Systems Principles (SOSP)*, 47–62. ACM. https://doi.org/10.1145/3341301.3359630.

Massalin, Henry. 1987. "Superoptimizer: A Look at the Smallest Program." In *Proceedings of the Second International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*.

Nandi, Chandrakana, Max Willsey, Amy Zhu, Yisu Remy Wang, Brett Saiki, Adam Anderson, Adriana Schulz, Dan Grossman, and Zachary Tatlock. 2021. "Rewrite Rule Inference Using Equality Saturation." *Proceedings of the ACM on Programming Languages* 5 (OOPSLA): 1–28. https://doi.org/10.1145/3485496.

Pal, Anjali, Brett Saiki, Ryan Tjoa, Cynthia Richey, Amy Zhu, Oliver Flatt, Max Willsey, Zachary Tatlock, and Chandrakana Nandi. 2023. "Equality Saturation Theory Exploration à La Carte." *Proceedings of the ACM on Programming Languages* 7 (OOPSLA2). https://doi.org/10.1145/3622834.

Silver, David, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, et al. 2017. "Mastering the Game of Go Without Human Knowledge." *Nature* 550: 354–59.

Singh, Zak. 2022. "Deep Reinforcement Learning for Equality Saturation." Master's thesis, University of Cambridge.

Smith, Gus Henry, Andrew Liu, Steven Lyubomirsky, Scott Davidson, Joseph McMahan, Michael Taylor, Luis Ceze, and Zachary Tatlock. 2021. "Pure Tensor Program Rewriting via Access Patterns." In *Proceedings of Machine Learning and Systems (MLSys)*, 3:273–86.

Tate, Ross, Michael Stepp, Zachary Tatlock, and Sorin Lerner. 2009. "Equality Saturation: A New

Approach to Optimization." In *Proceedings of the 36th ACM SIGPLAN-SIGACT Symposium on Principles of Programming Languages (POPL)*, 264–76. ACM. https://doi.org/10.1145/1480881.1480915.

Thomas, Samuel, and James Bornholt. 2024. "Automatic Generation of Vectorizing Compilers for Customizable Digital Signal Processors." In *Proceedings of the 29th ACM International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 19–34. ACM. https://doi.org/10.1145/3617232.3624873.

Trinh, Trieu H., Yuhuai Wu, Quoc V. Le, He He, and Thang Luong. 2024. "Solving Olympiad Geometry Without Human Demonstrations." *Nature* 625: 476–82. https://doi.org/10.1038/s41586-023-06747-5.

Willsey, Max, Chandrakana Nandi, Yisu Remy Wang, Oliver Flatt, Zachary Tatlock, and Pavel Panchekha. 2021. "Egg: Fast and Extensible Equality Saturation." *Proceedings of the ACM on Programming Languages* 5 (POPL): 1–29. https://doi.org/10.1145/3434304.

Yang, Yichen, Yisu Remy Wang, Max Willsey, Pavel Panchekha, Zachary Tatlock, et al. 2021. "Equality Saturation for Tensor Graph Superoptimization." In *Proceedings of Machine Learning and Systems (MLSys)*, 3:255–68.

Zhang, Yihong, Yisu Remy Wang, Max Willsey, and Zachary Tatlock. 2023. "Better Together: Unifying Datalog and Equality Saturation." *Proceedings of the ACM on Programming Languages* 7 (PLDI): 468–92. https://doi.org/10.1145/3591239.